COS40007 Artificial Intelligence for Engineering

Week 5 Studio Activities

ILO	Understand Deep Learning using TensorFlow and Kears.
Aim	• Learn what is CNN (Convolutional Neural Network) • Learn how to label object image data using the labelling tool • Learn how to train a CNN model using labelled data and use it for classification and Object detection
Resources	Books: 1. Prosise, Jeff. Applied machine learning and AI for engineers. " O'Reilly Media, Inc.", 2022. 2. Raschka, Sebastian, Yuxi Hayden Liu, and Vahid Mirjalili. Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python. Packt Publishing Ltd, 2022. Web Resources: 1. https://www.geeksforgeeks.org/convolutional-neural-network- cnn-in-machine-learning/ 2. https://github.com/labelmeai/labelme 3. https://www.geeksforgeeks.org/residual-networks-resnet- deep-learning/
Requirements for to be marked as complete	Demonstrate the data you labelled to your tutor to access your data labelling quality.

Note: It is recommended that students use Google Collab or other free GPU computing environments to complete activities 2-4 of this studio.

Studio Activity 1: Class labelling

In this studio activity, we will learn how to generate ground truth information by annotating our desired class. We will learn how to annotate image data at

1) Image level so your classification algorithm will classify the entire image
2) Object level so your classification algorithm will identify your targeted object (e.g., car, people) in the image file

Rust dataset

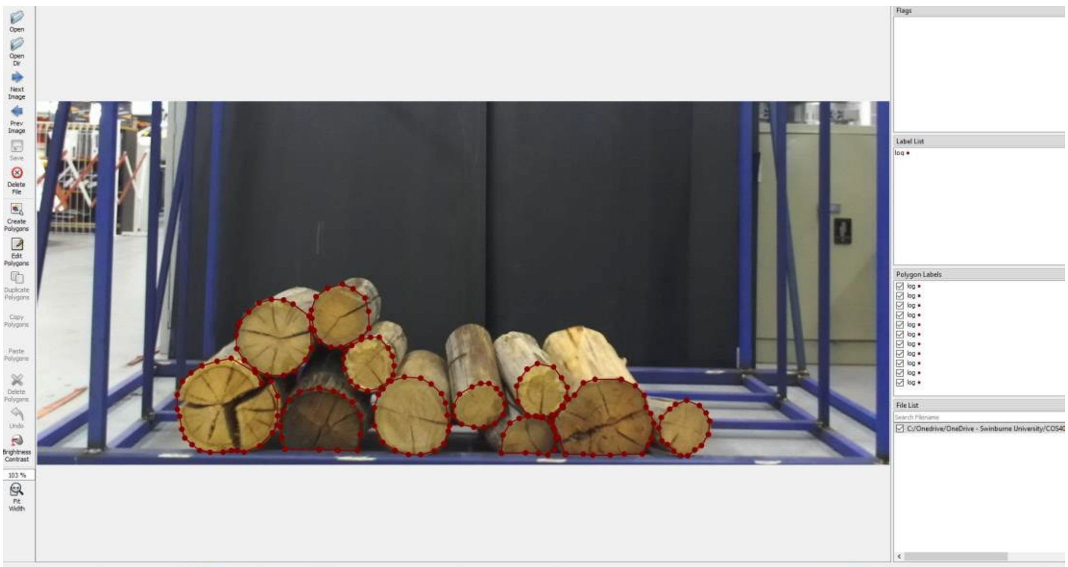
- Our image classification problem aims to identify whether the structural asset image contains rust (corrosion) or no rust.
- The image dataset of this is located here: Corrosion-dataset. However, data are not labelled (rust/ no rust)
- Some examples of labelling or annotation of rust and no-rust are provided here: Corrosion.
- Based on the example labels, you are required to label (or it is called an annotation) the data located in not labelled.
- Randomly label 10 images that are not labelled images by placing them in 2 folders. 'rust' or 'no rust'.

Log dataset

As part of your Studio 1 activity, you should install labelme on your machine. It is a Python-based image annotation tool. More details are here.

The image dataset for the wooden log is located in log-images. A sample labelme annotation can be found in the log-labelled. You can copy this folder to your local machine.

- To open the labelme GUI from your command line, type labelme.
- Now open the folder log-labelled from labelme (using open dir). You should see the wooden log's annotation for one image and how it is done using polygonal annotation.



- Now, you will need to perform a similar annotation task of wooden logs for the images located in log-images.
- Randomly pick five images and annotate all the logs in those five images using polygonal annotation. Get help from your tutor if you are unsure how to annotate.

Studio Activity 2: Develop your custom CNN for image classification

Follow this tutorial to understand how to develop your own CNN model to train a test dataset. For this, *minst* dataset is used. Try to create and run this code on your environment.

Studio Activity 3: Transfer Learning with RestNet for image classification

Follow this tutorial to understand how to use a pre-trained model to train with your data and build a deep learning mode. Cat and dog images are used for classification in this dataset. Try to create and run this code on your environment.

Studio Activity 4: Recursive CNN for object detection

Follow this tutorial to understand how to use a CNN for object recognition using a pre-trained recursive CNN model. For this dataset, kangaroo images are used as target objects. Try to create and run this code on your environment.

Next Steps:
In this studio, you have learned how to label image data and how you can use different deep neural networks for image classification and object recognition. In the portfolio assessment task, you will use the type of data that we labelled in this studio to develop CNN for image classification (in our case, rust and no rust for image classification and wooden logs for object recognition). Please follow the specifications for the portfolio task. The assessment Task for Week 5 can now be attempted and submitted via Canvas.